



Chapter 5

Gramene: A Resource for Comparative Analysis of Plants Genomes and Pathways

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Abstract

Gramene is an integrated bioinformatics resource for accessing, visualizing, and comparing plant genomes and biological pathways. Originally targeting grasses, Gramene has grown to host annotations for over 90 plant genomes including agronomically important cereals (e.g., maize, sorghum, wheat, teff), fruits and vegetables (e.g., apple, watermelon, clementine, tomato, cassava), specialty crops (e.g., coffee, olive tree, pistachio, almond), and plants of special or emerging interest (e.g., cotton, tobacco, cannabis, or hemp). For some species, the resource includes multiple varieties of the same species, which has paved the road for the creation of species-specific pan-genome browsers. The resource also features plant research models, including *Arabidopsis* and C4 warm-season grasses and brassicas, as well as other species that fill phylogenetic gaps for plant evolution studies. Its strength derives from the application of a phylogenetic framework for genome comparison and the use of ontologies to integrate structural and functional annotation data. This chapter outlines system requirements for end-users and database hosting, data types and basic navigation within Gramene, and provides examples of how to (1) explore Gramene's search results, (2) explore gene-centric comparative genomics data visualizations in Gramene, and (3) explore genetic variation associated with a gene locus. This is the first publication describing in detail Gramene's integrated search interface—intended to provide a simplified entry portal for the resource's main data categories (genomic location, phylogeny, gene expression, pathways, and external references) to the most complete and up-to-date set of plant genome and pathway annotations.

Key words Genomes, Comparative genomics, Phylogenetics, Gene trees, Gene homology, Synteny, Genetic variation, Structural variation, Pathways, Orthologs, Paralogs

1 Introduction

Gramene (<http://www.gramene.org>) is a plant phylogenomic and pathways resource built upon best-of-class open source software including Ensembl, Reactome, and Expression Atlas. It was designed to perform comparative functional genomic analysis and visualizations in crops and model plant species.

This chapter summarizes updates to the Gramene website and database since the last edition of this volume [1]. The current

Gramene build, #63 (October 2020), provides information on 93 reference genomes (*see* Table 1) including nearly four million genes, 122,947 gene family trees, 382 pairwise DNA alignments, and 84 synteny maps. This represents an almost threefold increase in genomics data relative to the previous edition, which included 32 genomes. Important updates were performed on existing assemblies and gene annotation sets of agronomically important species, including maize, sorghum, and bread wheat. The genes and trees provide a framework for transferring knowledge about genetic markers, QTLs of agronomic traits, and conserved gene functions between species. Orthologous genes support propagation of gene function, provide anchors for synteny, and reveal conservation of genomic location harboring agronomically important loci. Information on gene family expansion, as well as local duplications, provides insights into adaptive responses within and between species. In addition to providing anchors for transferring knowledge about genome function between species, protein-level alignments can be used as tools to identify artifacts in the structural annotations for a gene. Information about genetic and structural variation is provided for 15 species (*see* Table 2), including chemically induced EMS variants for sorghum and wheat. Since the previous update, Gramene (Fig. 1) and its pathways portal, the Plant Reactome, have been redesigned, whereas the Cyc pathway databases were decommissioned and archived. In addition, Gramene's and Plant Reactome's homepages were redesigned, and an advanced integrative and interactive search interface was implemented at Gramene's core.

We have also dedicated substantial efforts to evaluating and improving gene structural annotations, as accurate gene structure is necessary for assessment of gene function and critical for the application and adoption of gene editing approaches. In combination with traditional breeding efforts, such approaches could be used to build on future germplasm improvement strategies.

In this chapter, we have laid out three exercises, each with stepwise instructions, in which users can:

1. Explore Gramene's search results.
2. Explore gene-centric comparative genomics data visualizations in Gramene.
3. Explore genetic variation associated with a gene locus.

2 Materials

A computer with internet access and a standard web browser such as Mozilla/Firefox, Internet Explorer, Chrome, or Safari.

Table 1
List of plant reference genomes in Gramene (build 63, October 2020)

Species	Assembly	ENA accession	Literature references
<i>Actinidia chinensis</i> subsp. <i>chinensis</i> Red5	Red5_PS1_1.69.0	GCA_003024255.1	Pilkington et al. [2]
<i>Aegilops tauschii</i> subsp. <i>stragulata</i>	Aet_v4.0	GCA_002575655.1	Luo et al. [3]
<i>Amborella trichopoda</i>	AMTR1.0	GCA_000471905.1	Chamala et al. [4], Amborella Genome Project [5]
<i>Ananas comosus</i>	F153	GCA_902162155.1	Chen et al. [6]
<i>Arabidopsis halleri</i> subsp. <i>gemmifera</i>	Ahal2.2	GCA_900078215.1	Briskine et al. [7]
<i>Arabidopsis lyrata</i> subsp. <i>lyrata</i>	v.1.0	GCA_000004255.1	Hu et al. [8]
<i>Arabidopsis thaliana</i>	TAIR10	GCA_000001735.1	Cheng et al. [9]
<i>Arabis alpina</i>	A_alpina_V4	GCA_000733195.1	Willing et al. [10]
<i>Beta vulgaris</i> subsp. <i>vulgaris</i>	RefBeet-1.2.2	GCA_000511025.2	Dohm et al. [11]
<i>Brachypodium distachyon</i>	Brachypodium_distachyon_v3.0	GCA_000005505.4	Fox et al. [12]
<i>Brassica napus</i>	AST_PRJEB5043_v1	GCA_000751015.1	Chalhoub et al. [13]
<i>Brassica oleracea</i> subsp. <i>oleracea</i>	BOL	GCA_000695525.1	Parkin et al. [14]
<i>Brassica rapa</i>	Brapa_1.0	GCA_000309985.1	Wang et al. [15]
<i>Camelina sativa</i>	Cs	GCA_000633955.1	camelinadb.ca [16]
<i>Cannabis sativa</i>	cs10	GCA_900626175.1	Agricultural Genomics Foundation [17]
<i>Capsicum annuum</i>	ASM51225v2	GCA_000512255.2	Wang et al. [15]
<i>Chara braunii</i>	Cbr_1.0	GCA_003427395.1	Nishiyama et al. [18]
<i>Chlamydomonas reinhardtii</i>	Chlamydomonas_reinhardtii_v5.5	GCA_000002595.3	Merchant et al. [19]
<i>Chondrus crispus</i>	ASM35022v2	GCA_000350225.2	Janouškovec et al. [20]
<i>Citrullus lanatus</i>	Cla97_v1	GCA_000238415.2	Guo et al. [21]

(continued)

Table 1
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Species	Assembly	ENA accession	Literature references
<i>Citrus clementina</i>	Citrus_clementina_v1.0	GCA_000493195.1	Wu et al. [22]
<i>Coffea canephora</i>	AUK_PRJEB4211_v1	GCA_900059795.1	Denoeud et al. [23]
<i>Corchorus capsularis</i>	CCACVL1_1.0	GCA_001974805.1	Islam et al. [24]
<i>Cucumis melo</i>	Melonv4	GCA_902497455.1	Castanera et al. [25]
<i>Cucumis sativus</i>	ASM407v2	GCA_000004075.2	Huang et al. [26]
<i>Cyanidioschyzon merolae strain 10D</i>	ASM9120v1	GCA_000091205.1	Matsuzaki et al. [27]
<i>Cynara cardunculus</i> subsp. <i>scolymus</i>	CcrdV1	GCA_001531365.1	Scaglione et al. [28]
<i>Daucus carota</i> subsp. <i>sativus</i>	ASM162521v1	GCA_001625215.1	Iorizzo et al. [29]
<i>Dioscorea rotundata</i>	TDr96_F1_Pseudo_Chromosome_v1.0	GCA_002240015.2	Tamiru et al. [30]
<i>Eragrostis curvula</i>	CERZOS_E.curvula1.0	GCA_007726485.1	Carballo et al. [31]
<i>Eragrostis tef</i>	ASM97063v1	GCA_000970635.1	Cannarozzi et al. [32]
<i>Galdieria sulphuraria</i>	ASM34128v1	GCA_000341285.1	Schönknecht et al. [33]
<i>Glycine max</i>	Glycine_max_v2.1	GCA_000004515.4	Schmutz et al. [34]
<i>Gossypium raimondii</i>	Graimondii2_0	GCA_000327365.1	Paterson et al. [35]
<i>Helianthus annuus</i>	HanXRQr1.0	GCA_002127325.1	Badouin et al. [36]
<i>Hordeum vulgare</i> subsp. <i>Golden Promise</i>	GPv1	GCA_902500625.1	Schreiber et al. [37]
<i>Hordeum vulgare</i> subsp. <i>vulgare</i>	IBSC_v2	GCA_900075435.2	Mascher et al. [38]
<i>Ipomoea triloba</i>	ASM357664v1	GCA_003576645.1	Wu et al. [39]
<i>Leersia perrieri</i>	Lperr_V1.4	GCA_000325765.3	Stein et al. [40]
<i>Lupinus angustifolius</i>	LupAngTanjil_v1.0	GCA_001865875.1	Hane et al. [41]

(continued)

Table 1
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Species	Assembly	ENA accession	Literature references
<i>Malus domestica</i> <i>Golden</i>	ASM211411v1	GCA_002114115.1	Daccord et al. [42]
<i>Manihot</i> <i>esculenta</i>	Manihot_esculenta_v6	GCA_001659605.1	Prochnik et al. [43]
<i>Marchantia</i> <i>polymorpha</i>	Marchanta_polymorpha_v1	GCA_003032435.1	Bowman et al. [44]
<i>Medicago</i> <i>truncatula</i>	MedtrA17_4.0	GCA_000219495.2	Tang et al. [45]
<i>Musa</i> <i>acuminata</i> subsp. <i>malaccensis</i>	ASM31385v1	GCA_000313855.1	D'Hont et al. [46]
<i>Nicotiana</i> <i>attenuata</i>	NIATTr2	GCA_001879085.1	Xu et al. [47]
<i>Nymphaea</i> <i>colorata</i>	ASM883128v1	GCA_008831285.1	Zhang et al. [48]
<i>Olea europaea</i> subsp. <i>sylvestris</i>	O_europaea_v1	GCA_002742605.1	Unver et al. [49]
<i>Oryza barthii</i>	O.barthii_v1	GCA_000182155.2	Stein et al. [40]
<i>Oryza</i> <i>brachyantha</i>	Oryza_brachyantha.v1.4b	GCA_000231095.2	Stein et al. [40]
<i>Oryza</i> <i>glaberrima</i>	Oryza_glaberrima_V1	GCA_000147395.1	Stein et al. [40]
<i>Oryza</i> <i>glumacpatula</i>	Oryza_glumacpatula_v1.5	GCA_000576495.1	Stein et al. [40]
<i>Oryza longi</i> <i>staminata</i>	O_longistaminata_v1.0	GCA_000789195.1	Zhang et al. [50]
<i>Oryza</i> <i>meridionalis</i>	Oryza_meridionalis_v1.3	GCA_000338895.2	Stein et al. [40]
<i>Oryza nivara</i>	Oryza_nivara_v1.0	GCA_000576065.1	Stein et al. [40]
<i>Oryza punctata</i>	Oryza_punctata_v1.2	GCA_000573905.1	Stein et al. [40]
<i>Oryza rufipogon</i>	OR_W1943	GCA_000817225.1	Stein et al. [40]
<i>Oryza sativa</i> subsp. <i>indica</i>	ASM465v1	GCA_000004655.2	Stein et al. [40]
<i>Oryza sativa</i> subsp. <i>japonica</i>	IRGSP-1.0	GCA_001433935.1	Stein et al. [40]
<i>Ostreococcus</i> <i>lucimarinus</i> CCE9901	ASM9206v1	GCA_000092065.1	Palenik et al. [51]

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Table 1
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Species	Assembly	ENA accession	Literature references
<i>Panicum hallii</i> <i>FIL2</i>	PHallii_v3.1	GCA_002211085.2	Lovell et al. [52]
<i>Panicum hallii</i> subsp. <i>hallii</i> (<i>HAL2</i>)	PhalliiHAL_v2.1	GCA_003061485.1	Lovell et al. [52]
<i>Phaseolus</i> <i>vulgaris</i>	PhaVulg1_0	GCA_000499845.1	Vlasova et al. [53]
<i>Physcomitrella</i> <i>patens</i>	Phypa_V3	GCA_000002425.2	Lang et al. [54]
<i>Pistacia vera</i>	PisVer_v2	GCA_008641045.1	Zeng et al. [55]
<i>Populus</i> <i>trichocarpa</i>	Pop_tri_v3	GCA_000002775.3	Tuskan et al. [56]
<i>Prunus avium</i>	PAV_r1.0	GCA_002207925.1	Shirasawa et al. [57]
<i>Prunus dulcis</i>	ALMONDv2	GCA_902201215.1	Alioto et al. [58]
<i>Prunus persica</i>	Prunus_persica_NCBIV2	GCA_000346465.2	International Peach Genome Initiative et al. [59]
<i>Rosa chinensis</i>	RchiOBHm-V2	GCA_002994745.2	Raymond et al. [60]
<i>Saccharum</i> <i>spontaneum</i>	Sspon.HiC_chr_asm	GCA_003544955.1	Zhang et al. [61]
<i>Selaginella</i> <i>moellendorffii</i>	v1.0	GCA_000143415.1	Banks et al. [62]
<i>Setaria italica</i>	Setaria_italica_v2.0	GCA_000263155.2	Bennetzen et al. [63], Zhang et al. [64]
<i>Setaria viridis</i>	Setaria_viridis_v2.0	GCA_005286985.1	Mamidi et al. [65]
<i>Solanum</i> <i>lycopersicum</i>	SL3.0	GCA_000188115.3	Tomato Genome Consortium [66]
<i>Solanum</i> <i>tuberosum</i>	SolTub_3.0	GCA_000226075.1	Potato Genome Sequencing Consortium et al. [67]
<i>Sorghum bicolor</i>	Sorghum_bicolor_NCBIV3	GCA_000003195.3	Paterson et al. [68], Mace et al. [69]
<i>Theobroma cacao</i> subsp. <i>Belizeian</i> <i>criollo</i>	Criollo_cocoa_genome_V2	GCA_000208745.2	Argout et al. [70], Cocoa Genome Hub [71]
<i>Theobroma cacao</i> subsp. <i>Matina</i> 1-6	Theobroma_cacao_20110822	GCA_000403535.1	Argout et al. [72], Cacao Genome Database [73]
<i>Trifolium</i> <i>pratense</i>	Trpr	GCA_900079335.1	Vega et al. [74]

(continued)

Table 1
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Species	Assembly	ENA accession	Literature references
<i>Triticum aestivum</i>	IWGSC RefSeq v1.0	GCA_900519105.1	International Wheat Genome Sequencing Consortium (IWGSC) et al. [75]
<i>Triticum aestivum</i> subsp. <i>Cadenza</i>	EIv1.1	GCA_902810645.1	Designing Future Wheat [76]
<i>Triticum aestivum</i> subsp. <i>Claire</i>	EIv1.1	GCA_902810655.1	Designing Future Wheat [76]
<i>Triticum aestivum</i> subsp. <i>Paragon</i>	EIv1.1	GCA_902810665.1	Designing Future Wheat [76]
<i>Triticum aestivum</i> subsp. <i>Robigus</i>	EIv1.1	GCA_902810685.1	Designing Future Wheat [76]
<i>Triticum aestivum</i> subsp. <i>Weebill</i>	WeebillV1	GCA_902810675.1	Designing Future Wheat [76]
<i>Triticum dicoccoides</i>	WEWSeq_v.1.0	GCA_002162155.1	Avni et al. [77]
<i>Triticum turgidum</i> subsp. <i>durum</i>	Svevo.v1	GCA_900231445	Maccaferri et al. [78]
<i>Triticum urartu</i>	ASM34745v1	GCA_000347455.1	Ling et al. [79]
<i>Vigna angularis</i>	Vigan1.1	GCA_001190045.1	Kang et al. [80]
<i>Vigna radiata</i> subsp. <i>radiata</i>	Vradiata_ver6	GCA_000741045.2	Kang et al. [81]
<i>Vitis vinifera</i>	12X	GCA_000003745.2	Jaillon et al. [82]
<i>Zea mays</i>	B73_RefGen_v4	GCA_000005005.6	Jiao et al. [83]

Table 2
Genetic and structural variation in Gramene (build 63, October 2020)

Species	# of SNPs and indels	# of structural variants	# EMS-induced mutations
<i>Arabidopsis thaliana</i>	12,883,854		
<i>Brachypodium distachyon</i>	327,988		
<i>Helianthus annuus</i>	11,834		
<i>Hordeum vulgare</i> subsp. <i>vulgare</i>	18,331,939		
<i>Malus domestica</i>	10,627,081		
<i>Oryza glaberrima</i>	7,704,409		
<i>Oryza glumacpatula</i>	4,901,344		
<i>Oryza sativa</i> Indica Group	5,049,329		
<i>Oryza sativa</i> Japonica Group	28,179,246	1278	
<i>Solanum lycopersicum</i>	71,103,414		
<i>Sorghum bicolor</i>	8,187,272	27,884	1,481,300
<i>Triticum aestivum</i>	18,093,911		13,727,635
<i>Triticum turgidum</i> subsp. <i>durum</i>	1,899,698		
<i>Vitis vinifera</i>	457,245		
<i>Zea mays</i>	50,114,708		



3977445 genes in 93 genomes

Gramene Portals



Genome Browser
Browse genomes with annotations, variation and comparative tools



Plant Reactome
Browse and analyze metabolic and regulatory pathways



Tools
Tools for processing both our data and yours



Plant Expression ATLAS
Browse plant expression results at EBI ATLAS



BLAST
Query our genomes with a DNA or protein sequence



Gramene Mart
An advanced query interface powered by BioMart



Track Hub Registry
A global centralised collection of publicly accessible track hubs



Outreach and Training
Educational resources and webinars



Bulk Downloads
FTP download of our data



Archive
Legacy tools and data (markers, Cyc pathways, etc)

Latest News

COST IntGrape Training (Nov 30-Dec 4, 2020) - Apply by Oct 31
Tue, 20 Oct 2020

Announcing Gramene Database release #63 with 26 new plant genomes!
Thu, 01 Oct 2020

The fascinating origin of some favorite foods and how plants help in the fight against COVID-19
Sat, 20 Jun 2020

Biocuration of Plant Developmental Processes in the Plant Reactome pathway resource
Tue, 26 May 2020

First Virtual Maize Annotation Jamboree (March 10-12th, 2020)
Mon, 18 May 2020

New and updated plant genomes and pathways datasets planned for the Gramene release #63 (July 2020)
Wed, 13 May 2020

PAG meeting 2020 notes and a user survey
Wed, 22 Jan 2020

Gramene at Plant & Animal Genome 2020
Thu, 09 Jan 2020

Teaching Scholarships! Apply by Dec 2 - Maize Annotation Jamboree - 2020 Maize Genetics Conference
Mon, 18 Nov 2019

Fig. 1 Gramene's homepage (<http://www.gramene.org>)

2.1 Hardware and Software Requirements for Users

2.2 Gramene System Components

Gramene is a web-based application that allows users to search and view biological data, making use of the Ensembl infrastructure for generating, distributing, and visualizing genome annotations [84], the Reactome data model and framework [85], Java-based widgets for Plant Reactome [86], and Expression Atlas data visualizations [87]. Data is maintained in distinct relational databases (MySQL), and users connect to the site using a standard web browser. User queries for static (HTML) and dynamic content are negotiated by the Apache webserver via the Solr search platform and a middleware layer written in Perl and JavaScript. Bulk downloads of data are provided through FTP sites at Gramene and Ensembl Genomes.

2.3 Local Installation of Gramene

The minimum hardware configuration required for a local installation of Gramene consists of a desktop or server with a multicore CPU, 16 GB of memory, and 2 TB of disk space. Installation inside a virtual machine is possible. A recent release of Linux, such as one from RedHat/CentOS or Ubuntu, is required (*see Note 1*). Required software packages include Apache web server, Perl, PHP, MySQL, OpenJava, Drupal, MongoDB, node.js, Redis, and Apache Solr (*see Note 2*). Solr can be downloaded from the Apache Solr website, and many of the others can be installed via the distribution package management system (yum, apt-get), *see Note 3*. For specific installation instructions of Ensembl and Reactome software, contact Gramene developers at feedback@gramene.org.

3 Methods

3.1 Basic Navigation of the Gramene Website

Gramene continues to be powered by multiple modular platforms. The main entry point for the system is the web homepage (<http://www.gramene.org>; Fig. 1). Every Gramene page contains a main navigation bar and module-specific navigation bars if applicable, a simple search form that can be refined to interrogate different species and data modules, a link to the homepage, and a link to our Contact/Feedback page (<http://www.gramene.org/feedback>). The Contact/Feedback link is set up to provide a feedback page; the URL from the page that the user was viewing at the time of attempted contact is automatically included in the message. The main navigation bar of the Gramene's browser genome is located at the top of each content page and provides entry points to Gramene's most commonly used analysis tools, a direct link for users to

upload their own data, another to download Gramene's data in bulk, and links to navigate the site, and the contact Gramene's helpdesk. Links to information about the project (About), site navigation (Help), and citing the resource (Cite) are provided at the bottom of each page. In addition to providing access points to Gramene's data modules and tools, Gramene's homepage (Fig. 1) also points to legacy data and resources (i.e., Gramene's Archives and Cyc pathway databases), and an outreach section with events, educational materials, and social media (including Gramene's YouTube channel).

3.2 Example Uses of Gramene

Short example videos demonstrating the use examples described below are available on Gramene's YouTube channel (<https://goo.gl/ln9RLD>) and Outreach page (www.gramene.org/outreach; see **Note 4**). For the exercises described below, we will use the sorghum *MSD2* gene, identified from a sorghum EMS-induced population. Loss of function in this gene increases fertility and grain number relative to wild-type sorghum [88].

Here, we will demonstrate the power of comparative genomics by describing ways to explore the results of a simple search in Gramene while encouraging users to pose adventurous questions and discover ingenious ways to answer them. For example, given a gene with a compelling phenotype in one species, the user can find out whether it is conserved in another plant species of interest, what functions are predicted for its protein product based on its homology to known proteins, the tissues in which it is expressed, the pathways it might be involved in, whether it has genetic variants, and whether there are mutagenized lines in which to look for a similar phenotype. More advanced users might want to take advantage of Gramene's powerful comparative genomics tools to address evolutionary questions such as: Is my gene plant-specific or ancestrally conserved among eukaryotes? Has there been a gene family expansion in my species or a closely related species?

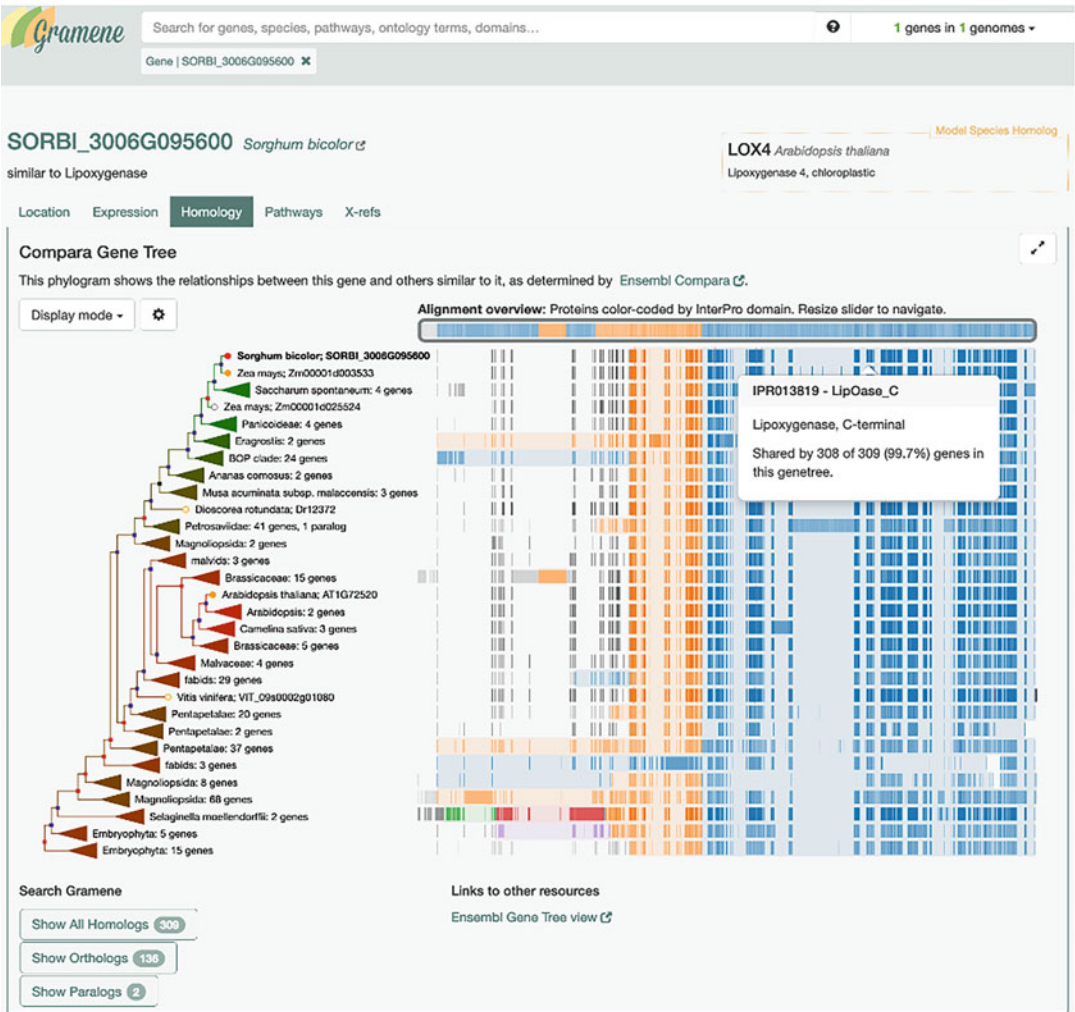
Exercise 1. Explore Gramene's Search Results

Gramene's integrated search interface was first described in 2018 [89], and more recently by Tello-Ruiz et al. [90]. The search engine is gene-centric, meaning that all available results correspond to features associated with a gene. Search results are grouped under the following categories: Location, Expression, Homology, Pathways, and Cross-References (Fig. 2a).

We will begin by searching for SORBI_3006G095600, the gene locus identifier for *MSD2* in Gladman et al. [88], in the search interface. Stepwise instructions for each exercise and a brief explanation of the observed outcomes are provided below. All images and data displayed in the interactive search results are downloadable.

1. Go to Gramene's homepage: www.gramene.org and locate the search box at the top of the page.
2. Type SORBI_3006G095600 in the search box. Select it from the list of suggested results (top and gene categories) that pops up. The default view for the results of queried genes that are part of a phylogenetic tree is the Homology view (Alignment overview display; Fig. 2a), which consists of the gene's family tree in its condensed form. We will discuss this view in more detail in the next exercise.
3. The gene's description ("similar to lipoyxygenase") appears under the gene identifier or gene name (*MSD2*). The closest homolog (LOX4—lipoyxygenase 4, chloroplastic) in a model species (in this case, *Arabidopsis thaliana*) is shown to the right of our queried gene. This information is very helpful for characterizing novel genes of unknown or putative function.
4. Observe the labels of the tabs for the five results categories mentioned above. Move your cursor to the Location tab (Fig. 2b), which will render a simplified image of the alternative transcripts for the gene of interest, in their genomic location. Exons are drawn as boxes joined by lines (introns); the coding bits or CDS are colored in red, and untranslated regions are colored in orange. At the bottom appear links to Gramene's Ensembl genome browser and external resources such as PhytoMine, an InterMine interface to data from Phytozome [91].
5. Switch to the Expression tab (Fig. 2c). This view depicts baseline gene expression data for a whole plant (top anatogram on the left of the results box) and for the plant's reproductive organs (bottom anatogram). Experiments are organized by study on the left side and by tissue or developmental stage on the top of the graph. Unique data points are colored in blue, with intensity increasing in proportion to the level of expression (i.e., higher expression data points are shown in darker blue). When you hover the cursor over a data point, the data point is highlighted in yellow, a more detailed summary pops up, and the corresponding tissue is colored in pink in the anatograms. If instead, you hover over a tissue on the anatogram on the left, the name of the tissue will pop up, and the points corresponding to expression data from this tissue will light up. This feature is the product of the collaboration between Gramene and EBI-Atlas.
6. Navigate to the Pathways tab (Fig. 2d). From the study by Gladman et al. [88], we know that MSD2 is part of the jasmonic acid (JA) biosynthesis pathway. Accordingly, the Pathways panel shows that this gene product catalyzes a reaction in two pathways, JA biosynthesis, and [13-LOX and 13-HPL], exactly as they appear in the Plant Reactome website. When you hover

A



B

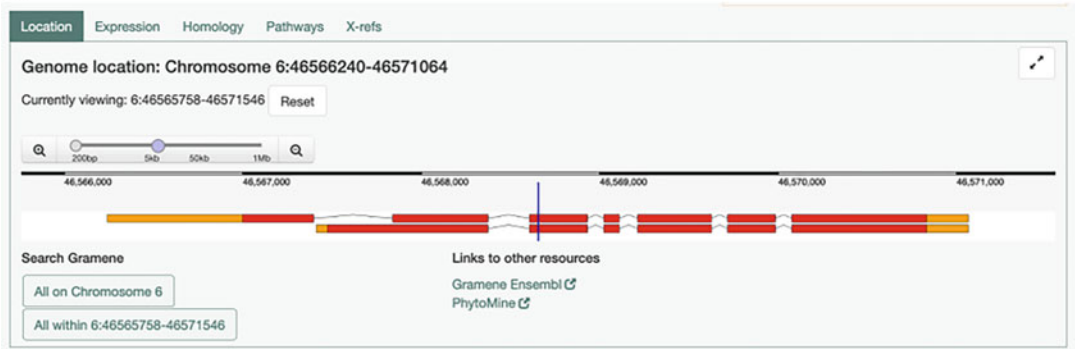


Fig. 2 Gramene's (gene-centric) search results. **(a)** Homology view (Alignment overview display). Condensed view of the MSD2 gene's family tree, and pop-out describing the color-coded (blue) C-terminal lipoxigenase domain. **(b).** Location view showing the gene's structure and alternative transcripts. **(c)** Baseline gene expression data for a whole plant (top anatogram), and its reproductive organs (bottom anatogram). **(d)** Pathways view. **(e)** Cross-references to external resources

C

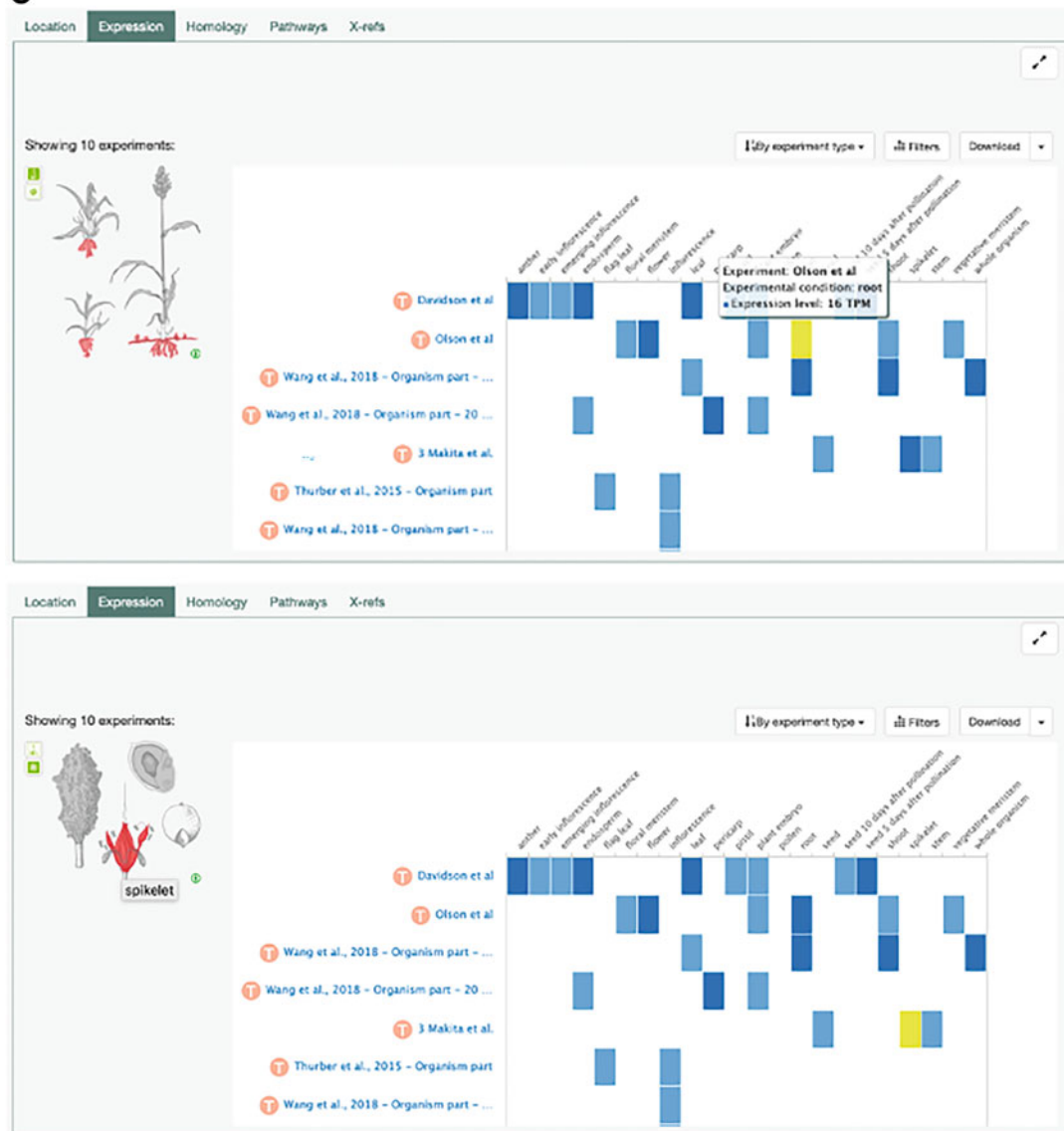
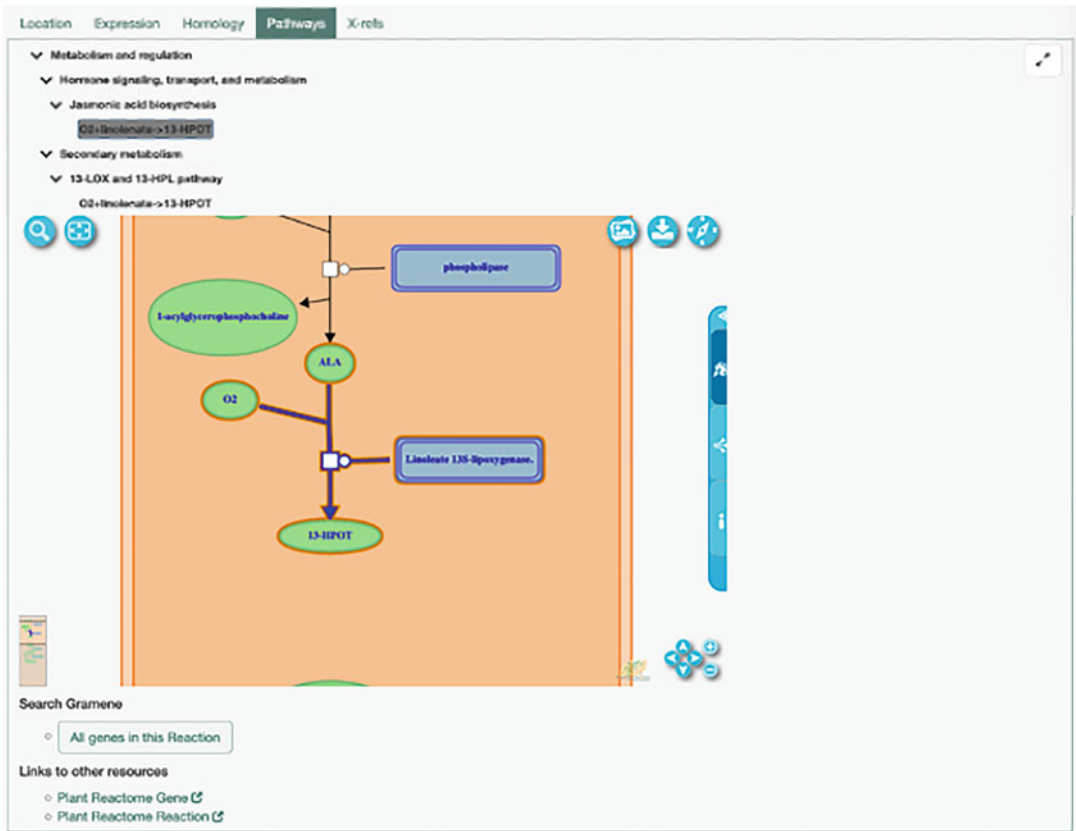


Fig. 2 (continued)

D



E

Database	IDs and links
UniProt	CSY999 ↗ C9DHL6 ↗
UniParc	UPI0001A86DA7 ↗
ENA	EES10882 ↗ EES10882.1 ↗
UniPathway	UPA00382 ↗
IntEnz	1.13.11.- ↗
UniGene	Sbl.387 ↗
QuickGO	CSY999 ↗

Fig. 2 (continued)

the cursor over the reaction in the hierarchy on the left side, the diagram on the right will show the reaction in either pathway context. Under the Search Gramene section, you can get a list of all genes in a given reaction or pathway using “All genes in this” You will also be able to navigate to the Gene and

corresponding Reaction pages in the Plant Reactome. There, you will be able to perform analyses of omics data, for example, overlaying gene expression data on pathways and generating animations to compare expression among tissues, among developmental stages, or between a mutant and the wild type. See, for example, a short video on overlaying omics data onto curated or projected pathways in Gramene's YouTube channel (*see Note 4*), where the animation shows changes in expression of genes in the JA biosynthesis pathway in four different scenarios: seed-bearing sessile spikelet tissue from the *msd2* mutant and wild type at developmental stages 4 and 5. There are various color schemes to choose from to show the changes in gene expression; in the animation described above, color changes from blue to yellow represent changes in expression from low to high. Pathway-based comparisons across species can be made using the Plant Reactome platform.

7. Navigate to the X-Refs tab (Fig. 2e). Here, you will find references to your gene in other databases, such as UniProt [92], ENA [93], QuickGO [94], etc.

Exercise 2. Explore Gene-Centric Comparative Genomics Data Visualizations in Gramene

In this exercise, we will explore comparative genomics data views available for the sorghum *MSD2* gene in Gramene. Loss-of-function point mutations in *MSD2* cause increased fertility and an increase in grain number relative to wild-type sorghum [88]. As before, we will begin by searching for SORBI_3006G095600, the gene locus identifier for *MSD2* from Gladman et al. [88], in the search interface.

1. By default, the condensed view for a gene is the Homology view (Alignment overview display), which consists of a phylogenetic tree in which the conserved protein domains are color-coded. Clicking on any of the colored boxes will prompt a pop-up inset describing the relevant InterPro domain [95] and its prevalence in the tree. All clades of a tree can be expanded or collapsed by clicking on the root nodes or condensed triangles of a collapsed tree branch.
2. Identify the rice ortholog of sorghum *MSD2*. Select "Pick Species from List" from the drop-down menu to the right of the search box (under the number of genes and genomes with a search hit). Select *Sorghum bicolor* and *Oryza sativa japonica*, and click "Done" to close the species window. Refresh your browser. You will now see only sorghum paralogs and rice orthologs of *MSD2* in the tree. Click on the "Show Orthologs" link to list the rice orthologs of *MSD2* (i.e., SPH/Os04g0447100 and Os05g0355800). Note that only

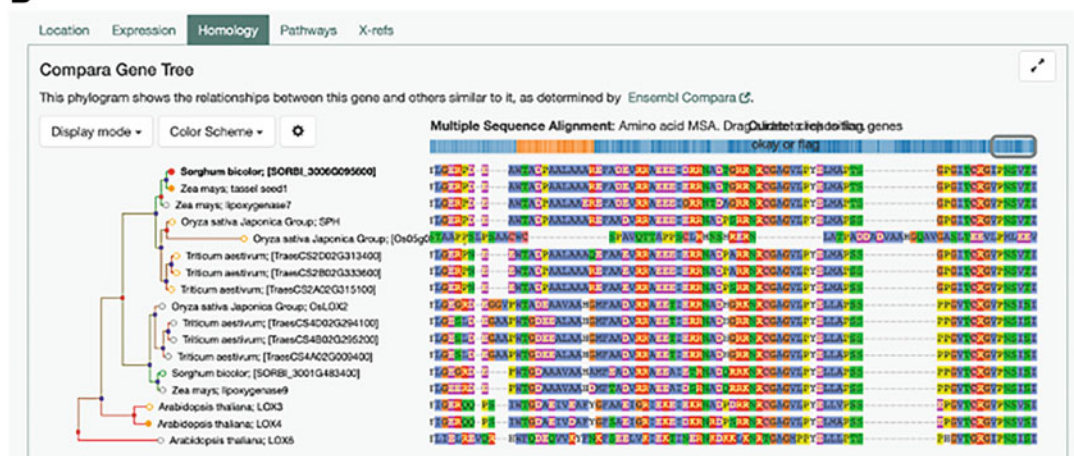
SPH is highly similar to *MSD2* and is thus the best rice ortholog (*see* **Note 4**).

3. Customize the species you want to be displayed in the phylogenetic tree. In the current build, #63, the *MSD2* tree includes 136 orthologous genes in 81 plant genomes (Fig. 2a). To decrease the complexity of the tree and show only a subset of the genomes, select the species you want to be displayed in your tree from the “Pick Species from List” option described above. The simplified tree shown in Fig. 3a–c was built by selecting sorghum, maize, rice, wheat, and *Arabidopsis*. Note: By selecting the “Show All Species” option in the drop-down menu, you will revert to the default tree with all species with orthologous genes in Gramene. Fully expanded trees with all species in Gramene (not shown) are useful to find pockets of microsynteny (for example, compare rice or wheat species among themselves, and relative to other clades), potential chromosomal inversions (see sugarcane genome components), and annotation errors (compare across the sugarcane genomes).
4. Explore the conservation of *MSD2* and the gene’s neighborhood using the scalable displays for the Homology results. The three possible views for the gene’s phylogeny are: Alignment overview, Multiple sequence alignment, and Neighborhood conservation. These are available from the “Display mode” drop-down menu on the left side of the Homology view. The default display, the Alignment overview, was described above. This view shows the entire *MSD2* protein aligned with its orthologs, and all known functional protein domains in contrasting colors. The Multiple sequence alignment display allows zooming to the amino acid sequence of the *MSD2* protein and traverses along it using the sliding bars in the overview or at the bottom of the tree. This view is particularly useful for detecting potential annotation errors or lineage-specific domains in regions that are highly conserved across species. Conversely, the Neighborhood conservation display allows users to compare entire gene neighborhoods; the gene being queried is depicted as a red line in the middle, and the ten genes flanking on each side are colored by family. This view provides insights into conservation of gene order by lining up potentially syntenic regions in closely related species, as well as determining the presence or absence of neighboring genes and structural variations. We will next zoom out this focused view to more interactive views of entire chromosomes in pairwise synteny maps.
5. Explore synteny for your gene of interest in other species and obtain lists of syntelogs in evolutionarily conserved regions. Synteny views provide insights into chromosome-level conservation of gene order and support access to defined lists of

A



B



C

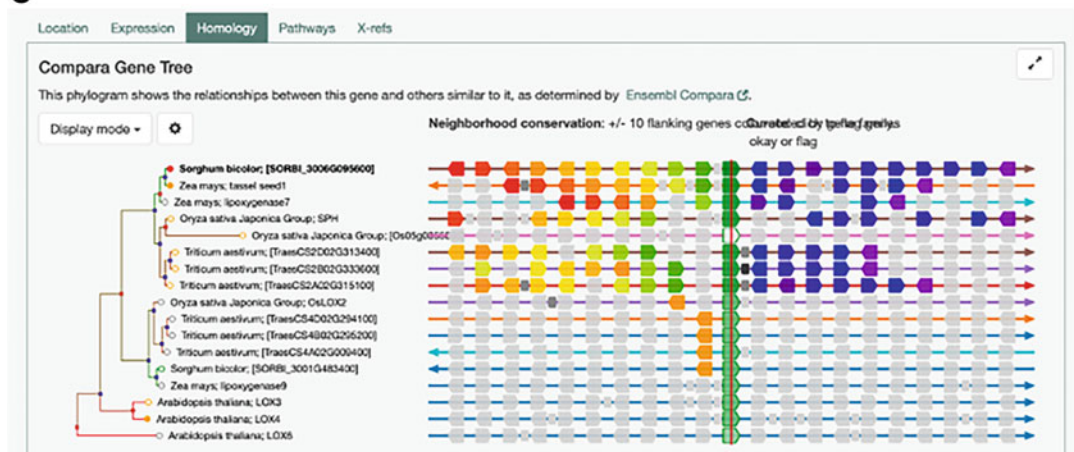
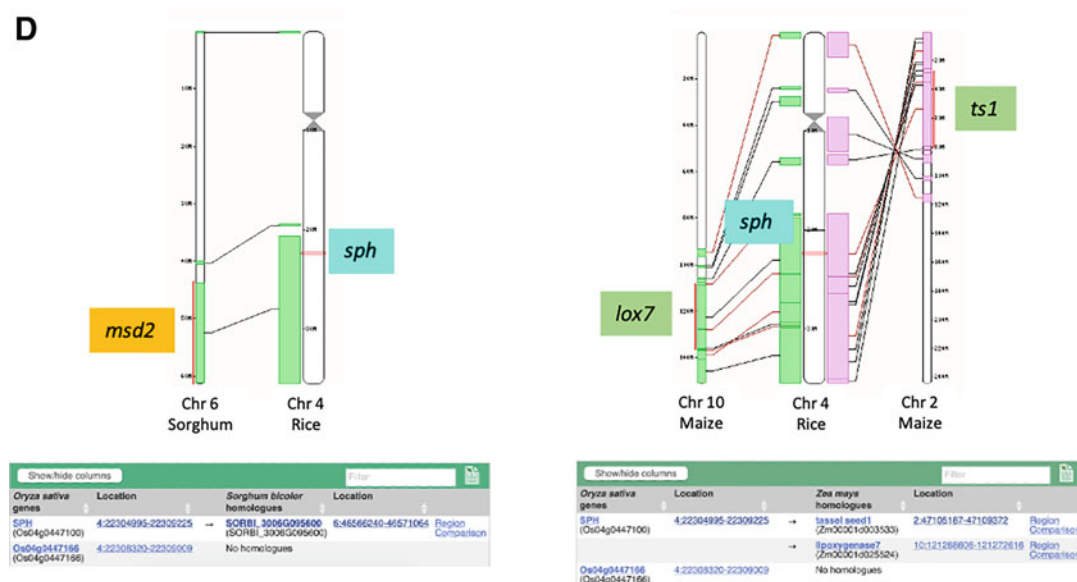


Fig. 3 Comparative genomics visualizations in Gramene. (a) Homology view with pruned tree (species shown: sorghum, maize, rice, wheat, and Arabidopsis). (b) Zoom to the amino acid level, showing the C-terminal end of the MSD2 protein. (c) Gene neighborhood view of the phylogenetic tree for sorghum MSD2. (d) Synteny map between sorghum and rice, and between rice and maize, each with a table listing the corresponding syntelogs in this region. (e) Region comparison view of the syntenic segment surrounding *MSD2*. (f) Gene gain/loss radial tree view. This circular view shows conservation of *MSD2* across species. By contrast, a sorghum drought-responsive gene (SORBI_3006G135500) that encodes galactose oxidase is absent in wheat, rice, maize, lovegrass, and sugarcane (grayed labels)

D



E

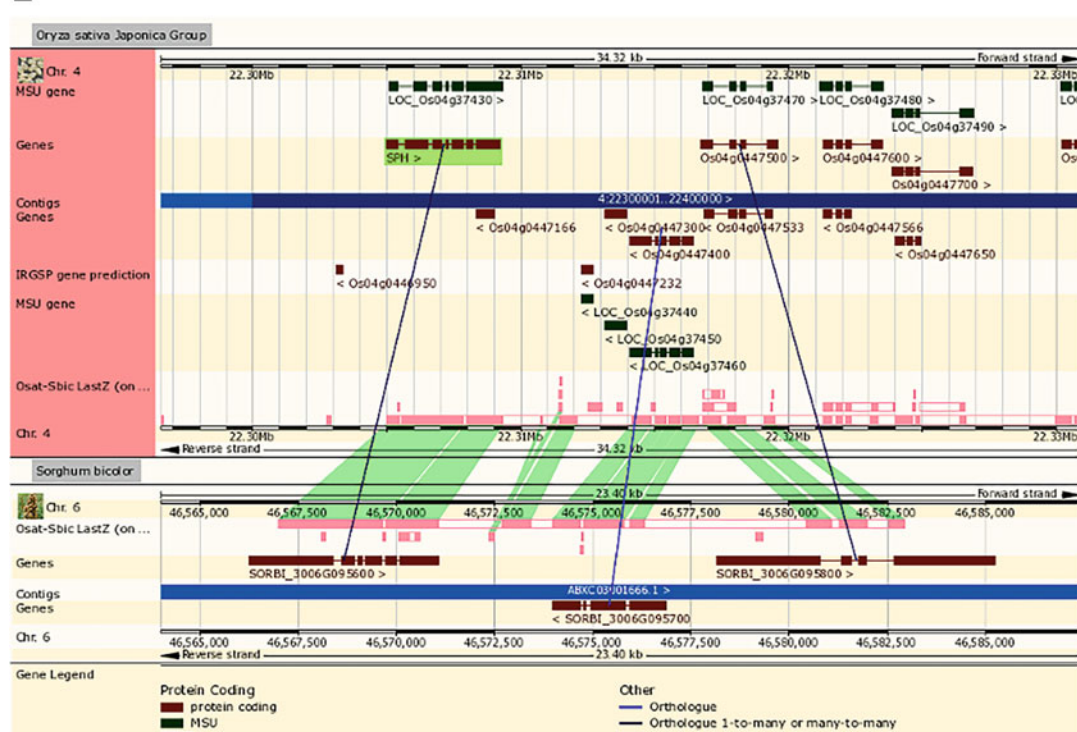


Fig. 3 (continued)

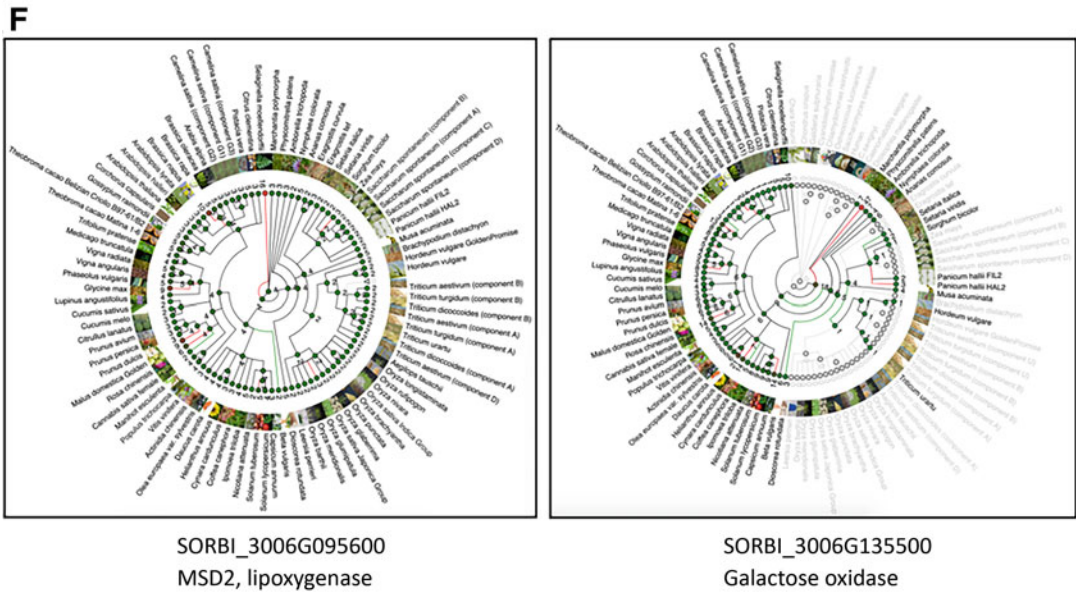


Fig. 3 (continued)

- orthologs. Synteny maps are available from the Location view of the Ensembl Genome Browser. To get to the Ensembl Location page for the *MSD2* gene region, use the link to its genomic location in the Location results box described in Exercise 1. Click on the Synteny option in the page's left side menu. In the synteny page, select the species that you want to compare. For this exercise, we will compare sorghum and rice and then rice and maize (Fig. 3d, e). We see that the sorghum *MSD2* gene is syntenic to its rice *SPH* ortholog. Synteny maps are particularly useful for visualizing genome duplications like the one that occurred in maize ~5–12 Mya followed by whole-genome rearrangement, which combined the duplicated chromosomes into a diploid genome [96]. Thus, in Fig. 3e, we see that rice *SPH* has two orthologs in maize: *TS1* and its paralog *LOX7*. Every synteny map comes with a table of syntelogs, each of which has a link to a Region Comparison view.
6. The Region Comparison view shows pairwise whole-genome DNA alignments. Figure 3f depicts the comparative view of the *MSD2* gene region between rice and sorghum.
 7. Another practical way to look at comparative aspects of a gene, and in particular whether a gene is present or absent in other species, are the gene gain/loss radial views. The circular view shows conservation of a gene across its phylogeny. In our sorghum *MSD2* example, the gene is conserved across the plant genomes represented in the tree. However there are cases where a gene is not present in some species, such as the

sorghum gene that encodes galactose oxidase (SORBI_3006G135500), which was recently shown to be responsive to drought [97]. As the radial gene tree shows, this gene is absent in wheat, rice, maize, lovegrass, and sugarcane, suggesting that it might be a rapidly evolving gene associated with an adaptive response. Although an apparent gene loss could be evolutionarily important for a species, it could also indicate a gap in the gene annotation, and this possibility should be further investigated.

Exercise 3. Explore Genetic Variation Associated with a Gene Locus

In Gramene, you can access a table containing genetic variants associated with a gene, including both naturally occurring and chemically induced variations (such as the single point mutations caused by ethyl methanesulfonate or EMS). Table 2 lists 15 plant species with genetic variants (i.e., SNPs, indels, and induced mutations), including two species with EMS-induced variants (sorghum and bread wheat) and two species with structural variation sets (sorghum and Japonica rice) in Gramene build #63. The variation summary tables and images (available for both genes and transcripts), as well as individual variant pages, were described in the previous edition of this chapter [1]. Herein, we present snapshots of those comprehensive resources (Figs. 4a–c) using sorghum *MSD2* as an example, before focusing on the newer sets of EMS-induced mutations in *MSD2*. Figure 4a shows a variation image for sorghum *MSD2*; Fig. 4b shows a variation table for one of the gene’s transcripts; and Fig. 4c presents an individual variant page listing available data (i.e., genomic context, sample genotypes, genotype frequency, phenotype data, LD display, etc.). Another useful variation table can be reached from a gene’s transcript page by selecting the “Variants” option under the “Protein Information” section (Fig. 4d). To limit the list of variants to EMS mutations, we used “EMS” as a criteria to filter the list shown in Fig. 4d. Note that all missense variants (including EMS mutations) in the variation tables include SIFT scores [98], which predict whether an amino acid substitution affects a protein’s function. Scores less than 0.05, shown in red, indicate that the corresponding substitution is predicted to be deleterious, whereas larger SIFT scores shown in blue or green indicate that a substitution is likely to be tolerated. To identify a sorghum (mutant) line that bears a deleterious EMS-induced mutation, you can reorder the variants in your table by ascending SIFT score and click on the name of a variant with SIFT <0.05 (red). Once in the individual variant’s page, go to the “Sample genotypes” panel and reorder the table based on the “Genotype” column to identify a sorghum accession (sample) with

a non-wild-type allele (EMS mutation), as shown in Fig. 4c (Sample genotypes inset). Alternatively, use the BioMart [99] utility to identify mutagenized lines with deleterious functional consequences in the MSD2 protein by using a combination of Filters such as “Sorghum_EMS_mutants” (source or population), and “stop_gained” or “protein_altering_variant” (variant consequence), and then including the samples with their corresponding genotypes (attributes) in your results (Fig. 4c). EMS-mutant lines will have a genotype that differs from the reference alleles at that position. To download all genetic variants for the *MSD2* gene, users can go to the Variation table/image and download the data directly with the Download table/image icons (Figs. 4a, b), click on the “Export data” option on the left sidebar menu, or download directly using the BioMart utility as described above (Fig. 4e).

4 Notes

1. To install Gramene locally, you will need some familiarity working with Linux and its software packaging system. Basically, you must have a working LAMP system (LAMP = Linux Apache MariaDB PHP) before attempting the installation.
2. To download the packages not available through the Linux distribution itself, we suggest:
 - (a) Node.js: <https://nodejs.org/en/download/>
 - (b) Drupal: <https://www.drupal.org/download>
 - (c) MongoDB: <https://www.mongodb.com/try/download/community>
 - (d) Solr: <https://lucene.apache.org/solr/downloads.html>
 - (e) Redis: <https://redis.io/download>
3. For CentOS, the yum packages are: perl httpd mariadb java-1.8.0-openjdk php7 (requires software collections to be installed on CentOS 7 or RHEL 7).
4. A series of short videos with specific tasks described in this chapter’s exercises (e.g., “How to find an ortholog for my gene of interest”) is available from Gramene’s YouTube channel at <https://goo.gl/ln9RLD>.

Acknowledgements

The authors thank all members of the Gramene Project and funding sources. We are also grateful to Gramene’s users for valuable suggestions and our collaborators for sharing genomic-scale

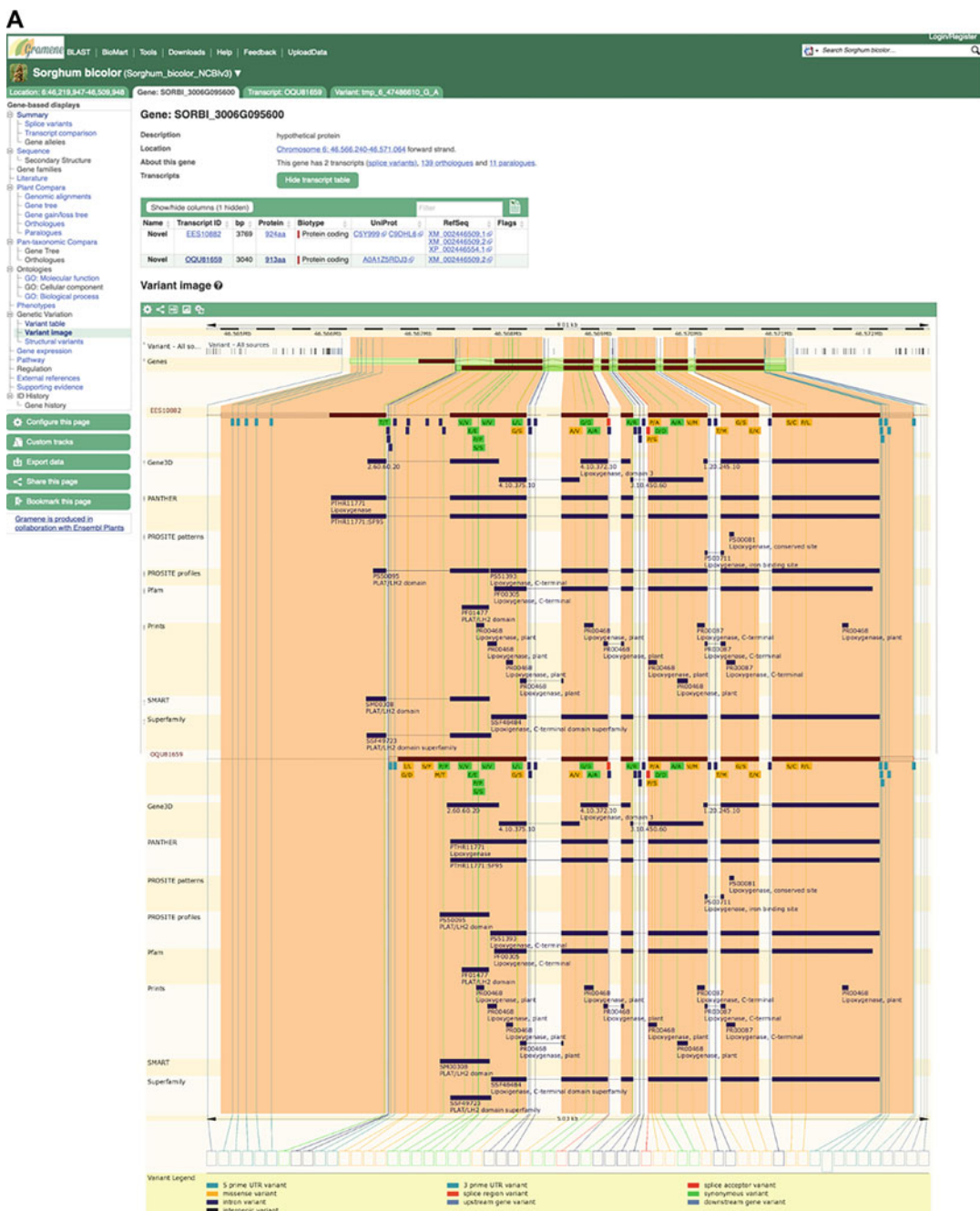


Fig. 4 Exploring genetic variation in Gramene. (a) Gene variant image for *MSD2*. Color-coded gene variants are depicted in the context of protein domains. Single letters are used to characterize amino acid substitutions in missense and EMS-induced variants. (b) Table of short variants for the shorter transcript of sorghum *MSD2* (OQU81659). The inset shows a blow-up of the variant page of the EMS-induced point mutation tmp_6_47486610_G_A, selected because a SIFT score <0.05 is predictive of a deleterious missense substitution (Gly290Ser in this case) in the *MSD2* protein. (c) Data displays for a sorghum EMS variant. To explore the available data for this variant, navigate through the options on the left menu bar, which correspond to the icons under the “Explore this variant” section. Data is hyperlinked to the boxes shown in the connecting insets: Genomic context (Genes and Regulation, and Flanking sequence), Sample genotypes, and Genotype frequency. (d) The variants table of a transcript includes SIFT scores as well as descriptive changes in codons and corresponding amino acids. This image shows a selection of EMS-induced mutations. (e) BioMart query for sorghum lines bearing deleterious EMS-induced mutations. Parameters used (filters) are shown in the left sidebar and top panels, attributes specified to be included in the results are displayed in the middle panel, and results are shown in the bottom panel. One example of such a mutant line is ARS133, with the genotype AIG; the reference allele at this position is G

B

BLAST | BioMart | Tools | Downloads | Help | Feedback | Update Data

Search Sorghum biot...

Location: 646,219,947-46,509,948 Gene: SORBI_000G095600 Transcript: OQUB1659 Variant: tmp_6_47486610_G_A

Transcript-based displays

- Summary
- Sequence
- Exons
- CDNA
- Protein
- Protein information
- Protein summary
- Domains & features
- Protein variants
- 3D Protein model
- Genetic Variation
- Variant table
- Variant image
- Population comparison
- Comparison image
- External References
- General identifiers
- Oligo probes
- Supporting evidence
- ID history
- Transcript history
- Protein history

Transcript: OQUB1659

Description: hypothetical protein

Location: Chromosome 6: 46,567,412-48,571,064 forward strand.

About this transcript: This transcript has 6 exons, is annotated with 26 domains and features and is associated with 61 variant alleles.

Gene: This transcript is a product of gene SORBI_000G095600 [Hide transcript table](#)

Show/Hide columns (1 hidden)

Name	Transcript ID	bp	Protein	Biotype	Unifrut	RefSeq	Flags
Novel	EES10882	3769	324aa	Protein coding	CS1999.9 G0016.9	XM_02446029.1 (p)	
Novel	OQUB1659	3040	313aa	Protein coding	ASA12003.9	XP_02446034.1 (p)	

Variant table

Filter: SIFT: All Consequences: All Filter Other Columns

Variant ID	Chr: bp	Alleles	Class	Source	Conseq. Type	AA	AA coord	SIFT
tmp_6_47486602_G_C	646567382	G/C	SNP	Maize_2013	upstream gene variant	-	-	-
tmp_6_47486707_T_C	646567406	T/C	SNP	Maize_2013	upstream gene variant	-	-	-
tmp_6_47486710_G_C	646567409	G/C	SNP	Maize_2013	upstream gene variant	-	-	-
tmp_6_47486725_G_A	646567424	G/A	SNP	Maize_2013	5 prime UTR variant	-	-	-
tmp_6_47486755_G_C	646567454	G/C	SNP	Maize_2013	5 prime UTR variant	-	-	-
tmp_6_47486752_G_A	646567542	G/A	SNP	Sorghum_EMS_mutants	missense variant	G/D	23	0.26
tmp_6_47486848_A_C	646567547	A/C	SNP	Maize_2013	missense variant	I/L	25	1
tmp_6_47486981_C_T	646567560	C/T	SNP	Maize_2013	missense variant	S/F	69	0.02
tmp_6_47486921_T_C	646567773	T/C	SNP	Maize_2013	missense variant	M/T	100	0.04
tmp_6_47486996_T_A	646567795	T/A	SNP	Maize_2013	synonymous variant	P	107	-
tmp_6_47486943_G_C	646567942	G/C	SNP	Maize_2013	synonymous variant	V	166	-
tmp_6_47486920_G_A	646567999	G/A	SNP	Maize_2013	synonymous variant	E	175	-
tmp_6_47486936_G_C	646568035	G/C	SNP	Maize_2013	synonymous variant	P	187	-
tmp_6_47486932_G_C	646568038	G/C	SNP	Maize_2013	synonymous variant	S	188	-
tmp_6_47486905	646568104	G/T	SNP	Munira_2013	synonymous variant	V	210	-
tmp_6_47486909_C_T	646568308	C/T	SNP	Sorghum_EMS_mutants	synonymous variant	L	278	-
tmp_6_47486810_G_A	646568309	G/A	SNP	Sorghum_EMS_mutants	missense variant	G/S	279	0.01
tmp_6_47486930_C_T	646568389	C/T	SNP	Maize_2013	intron variant	-	-	-
tmp_6_47486938_A_T	646568397	A/T	SNP	Maize_2013	intron variant	-	-	-
tmp_6_47486928	646568424	A/G	SNP	Munira_2013	intron variant	-	-	-
tmp_6_47486938_G_T	646568497	G/T	SNP	Sorghum_EMS_mutants	intron variant	-	-	-
tmp_6_47486936_C_T	646568705	C/T	SNP	Sorghum_EMS_mutants	missense variant	A/V	333	0.04
tmp_6_47486982_C_A	646568781	C/A	SNP	Maize_2013	synonymous variant	G	368	-
tmp_6_47487133_A_G	646568832	A/G	SNP	Maize_2013	synonymous variant	A	375	-
tmp_6_47487237_G_T	646568936	G/T	SNP	Maize_2013	splice region variant	-	-	-

Location: 646,219,947-46,509,948 Gene: SORBI_000G095600 Transcript: OQUB1659 Variant: tmp_6_47486610_G_A

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Genotype frequency
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

tmp_6_47486610_G_A SNP

Most severe consequence: missense variant | See all predicted consequences

Alleles: G/A | Highest population MAF: < 0.01

Location: Chromosome 6: 46568309 (forward strand) | VCF: 6 46568309 tmp_6_47486610_G_A G A

HGVs names: This variant has 5 HGVS names - [Hide](#)

- 6p.46568309G>A
- EES10882.1:c.868G>A
- EES10882.1:p.Gly290Ser
- OQUB1659.1:c.835G>A
- OQUB1659.1:p.Gly279Ser

External Links: [Original source](#)

About this variant: A Sorghum Mutant Resource as an Efficient Platform for Gene Discovery in Grasses. Contact Dr. Zhanguo Xin for seeds. This variant overlaps 2 transcripts and has 241 sample genotypes.

Explore this variant

- Genomic context
- Genes and regulation
- Flanking sequence
- Genotype frequency
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations

Using the website

- Video: [Browsing SNPs and CNVs in Ensembl](#)
- Video: [Clp: Genome Variation](#)
- Video: [BioMart: Variation IDs to HGNC Symbols](#)
- Exercise: [Genomes and SNPs in Malaria](#)

Analysing your data

[VeP](#) Test your own variants with the Variant Effect Predictor

Reference materials

- [Variation Quick Reference card](#)

Fig. 4 (continued)

C

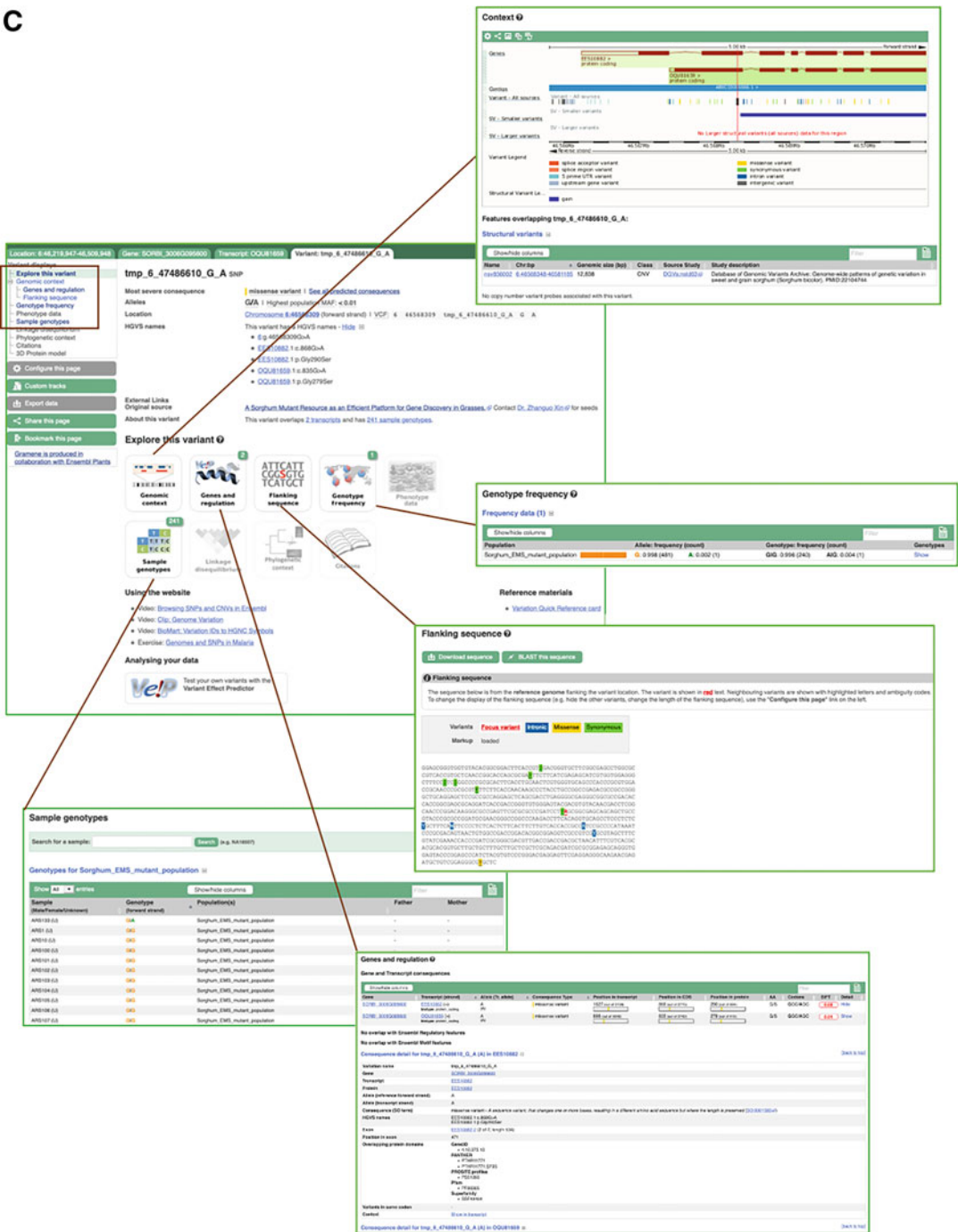


Fig. 4 (continued)

D

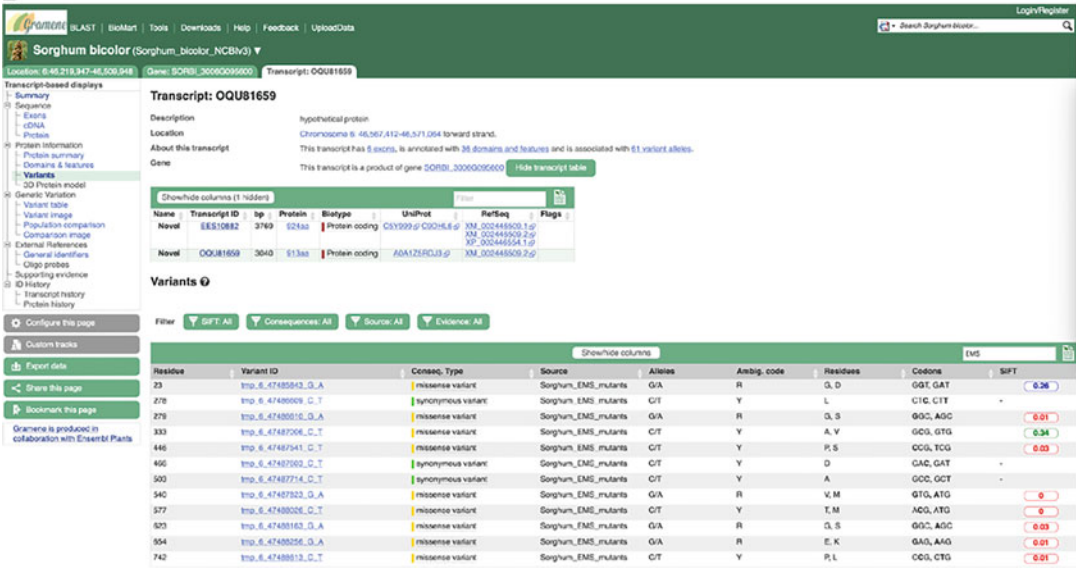


Fig. 4 (continued)

Dataset

Sorghum bicolor Short Variants (SNPs and indels excluding flagged variants) (Sorghum_bicolor_NCBIV3)

Filters

Gene stable ID(s) [Max 500 advised]; [ID-list specified]
Variant source : Sorghum_EMS_mutants
Population : Sorghum_EMS_mutant_population
Variant consequence : protein_altering_variant,stop_gained

Attributes

Variant name
Population name
Genotype frequency
Sample name
Sample genotype

Dataset

[None Selected]

Please restrict your query using criteria below

REGION:

GENERAL VARIANT FILTERS:

☒ Variant source

Variant source

Filter by Variant name (e.g. rs123, CM000001) [Max 500 advised]

☒ Population

Population

☐ Sample

Sample

☐ Global minor allele frequency <=

☐ Global minor allele frequency >=

☐ Variant supporting evidence

☐ Distance to transcript <=

☒ GENE ASSOCIATED VARIANT FILTERS:

☒ Gene stable ID(s) [Max 500 advised]

☒ Variant consequence

Variant source

Maize_2013
Maize_2013
Sorghum_EMS_mutants

Filter by Variant name (e.g. rs123, CM000001) [Max 500 advised]

Browse... No file selected.

Population

2,5p_lines
45,5p_lines
SAP
Sorghum_EMS_mutant_population

Sample

38432
JAI4
AKS1
AKS10
AKS100
AKS101
AKS102
AKS103
AKS104
AKS105
AKS106
AKS107
AKS108
AKS109
AKS11
AKS110
AKS111
AKS112
AKS113
AKS114

☐ Global minor allele frequency <=

☐ Global minor allele frequency >=

☐ Variant supporting evidence

☒ Only
☐ Excluded

☐ Distance to transcript <=

☒ GENE ASSOCIATED VARIANT FILTERS:

☒ Gene stable ID(s) [Max 500 advised]

☒ Variant consequence

Gene stable ID(s) [Max 500 advised]

Browse... No file selected.

Variant consequence

sequence_comparison
sequence_variant
splice_acceptor_variant
splice_donor_variant
splice_region_variant
splice_site_variant
splicing_variant
start_lost
stop_gained

Please select columns to be included in the output and hit 'Results' when ready

Variant ☒ Flanking Sequences

VARIANT ASSOCIATED INFORMATION:

Variant information

☒ Variant name
☐ Variant source
☐ Variant source description
☐ Chromosome/scaffold position start (bp)
☐ Chromosome/scaffold position end (bp)
☐ Strand
☐ Variant alleles

☐ Mapweight
☐ Variant supporting evidence
☐ Ancestral allele
☐ Minor allele (ALL)
☐ Global minor allele frequency (all individuals)
☐ Global minor allele count (all individuals)

Population Genetics

☒ Population name
☐ Population size

☐ Population genotype
☒ Genotype frequency

Sample

☒ Sample name

☒ Sample genotype

GENE ASSOCIATED INFORMATION:

Gene attribute

☐ Gene stable ID
☐ Transcript stable ID
☐ Transcript strand
☐ Biotype
☐ Variant consequence
☐ Consequence specific allele
☐ Protein allele

☐ Variant start in cDNA (bp)
☐ Variant end in cDNA (bp)
☐ Variant start in translation (aa)
☐ Variant end in translation (aa)
☐ Variant start in CDS (bp)
☐ Variant end in CDS (bp)
☐ Distance to transcript

Export all results to

File

Go

Email notification to

View

10

rows as

TSV

☒ Unique results only

Variant name	Population name	Genotype frequency	Sample name	Sample genotype
...				
tmp_6_47486610_G_A	Sorghum_EMS_mutant_population	0.00414938	ARS133	G A
tmp_6_47486610_G_A	Sorghum_EMS_mutant_population	0.00414938	ARS135	G G
tmp_6_47486610_G_A	Sorghum_EMS_mutant_population	0.00414938	ARS136	G G
tmp_6_47486610_G_A	Sorghum_EMS_mutant_population	0.00414938	ARS138	G G
tmp_6_47486610_G_A	Sorghum_EMS_mutant_population	0.00414938	ARS139	G G
...				

Fig. 4 (continued)

datasets that make Gramene an outstanding community resource. The Genomes and Pathways modules in Gramene would not have been possible without synergistic collaborations with the Ensembl Genomes project at the EMBL-European Bioinformatics Institute and the Reactome project at the Ontario Institute for Cancer Research, respectively.

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